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Contents

1 Introduction **3**
1.1 Project Background . . . 3
1.2 Project Overview . . . 7
1.2.1 Project Outline . . . 7
1.2.2 Project Objectives . . . 7
1.3 Project Hypothesis . . . 8

2 Literature and Technology Review **10**
2.1 Investigate existing mobile apps that use psychological methods to treat mental health issues . . . 10
2.1.1 Prior reviews on research-based mobile app interventions . . . 10
2.1.2 Conclusion . . . 12
2.2 Investigate the contextualisation of SA and SMA in existing apps . . . 12
2.2.1 Current contextualisation, or Past contextualisation issues (temp) 13
2.3 Investigate psychological methods used to treat smartphone and social media addiction . . . 14
2.3.1 section to be named . . . 14
2.4 Investigate methods to track app usage . . . 14
2.4.1 Justification for app usage tracking . . . 14
2.4.2 Methods of app usage tracking . . . 14
2.4.3 Discussion . . . 15

3 Project's Primary Research Methods **17**
3.1 Research Method and Justification . . . 17
3.2 Develop the Mobile app (temp) . . . 17
3.2.1 Development . . . 17
3.2.2 Project Lifecycle . . . 17
3.3 Usability Testing with users and Evaluation . . . 17
3.3.1 Usability Testing . . . 17
3.3.2 Evaluation . . . 17
3.4 Ethical Issues . . . 17
3.5 Preliminary Work . . . 17
3.6 Remaining Work . . . 17

References **18**

List of Figures

1 *Rules tab of AppDetox (Löchtefeld, Böhmer, and Ganev 2013)* . . . 6

2 *Violations tab of AppDetox (Löchtefeld, Böhmer, and Ganev 2013)* . 6

1 Introduction

This section will provide a brief overview on smartphones and social media. It will then provide context on the issue of smartphone addiction and social media addiction, including how people develop such addictions, and the effects they have on an individual. Existing apps and studies that attempt to deal with the issue are also mentioned. It will also provide an overview on the project and it's objectives to be achieved in the literature & technology review and during development of the app.

1.1 Project Background

Social media, sometimes referred to as Social Networking Sites (SNS), have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with groups of individuals online from around the globe on a never-before-seen scale, allowing for friendships, communities and relationships to be built from virtually anywhere. Dating back to the days of MySpace in the mid-2000s, to SNS's such as X (formerly known as Twitter) and Facebook reaching out to billions around the globe, social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of social media presence (Shah 2021).

Similarly, smartphones have taken the world by storm, and have also become an important part of life for most people, even at early ages, with studies reporting that young people are receiving their first phone at the age of 10 (Chan, Yeo, and Handayani 2023). With the June 2007 release of Apple's "iPhone" and the release of the HTC Dream running the open source "Android" operating system in 2008, both Apple's iPhone and the myriad of devices that use Google's Android OS have cultivated a large shift in the world of online media (Goggin 2012), with it being reported in 2021 that 85% of adults in the United States own a smartphone (Ouellette and Rao 2022). Smartphones nowadays are used for communication via calling or texting through apps such as WhatsApp, for entertainment (gaming, TV, online content, etc..), for productivity, for learning, and a wide variety of other purposes.

However, over the years, worries have been raised regarding a variety of issues that concern both smartphones and social media, particularly surrounding the problem of individuals developing an overdependence on their smartphones and / or social media that negatively affects their day-to-day life **put a NEW citation here**. Commonly categorised as "Smartphone Addiction" (SA) and "Social Media Addiction" (SMA), the issue has become a largely discussed and researched topic over the last 15 years with researchers looking into potential causes and correlations with

other health effects.

While SA isn't classified the same way other addictions such as drug addiction are classified, the nature of it is instead classified as a behavioural addiction (King and Dong 2017), and a similar manner, SMA is considered as a subset of Internet Addiction (Hou et al. 2019), which itself is considered a subset of behavioural addictions (Chou, Condron, and Belland 2005), and while both are typically categorised as separate types of "addiction", it has been argued by some that SA may be correlated with SMA, due to smartphones themselves being enablers for addictive behaviors, including SMA (Kuss and Griffiths 2017).

The nature of SA and SMA has on young individuals have been extensively studied over the years, including why individuals may develop an addiction to both in the first place, and the effects that are correlated with both. For example, when looking at why some may be addicted to smartphones and social media, a study from Indian researchers Bhalerao et al. (2020) that analysed SA provided a multitude of possible causes behind SA, one of which being how the current generation has been brought up alongside the internet and technology as a whole, therefore they may feel a preference to communicate over the internet as opposed to in-person, fuelling the dwindling need for talking face-to-face with someone. When looking at the effects both are correlated with, in 2017, researchers in the United States King and Dong (2017) made a series of hypotheses in relation to college students and the effects related to SA, in particular, they looked at SA's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and social well-being. The results of their study found that SA could be positively correlated with life stressors, digital overload, retention, and academic performance, with social well-being as the only factor with no significant correlation. Following the results, they concluded by mentioning that 3 out of their 5 hypotheses were consistent with past studies, while mentioning that regarding the hypothesis related to retention was "...a novel insight created from this study as current SA has not linked the consideration with college retention.".

Many researchers and developers have made attempts to study and provide treatment for the issue of SA and SMA, both as experimental studies on alternative treatments and as dedicated mobile applications. A 2024 India-based study (Seran 2024) aimed to look at using educational-based methods to reduce SMA and obesity. After answering a questionnaire, participants were split into a control group and a experimental group, the latter of which were then involved in educational activities such as lectures, brainstorming sessions, and exercises, among other activities. Participants

were then scored on their SMA one, three, and six months after the initial study and the researchers found that there was a 21.2% difference in SMA scores between the control group and the group that partook in the educational-based intervention, with them concluding that it was an effective measure in reducing SMA. A China-based study from 2019 analysing SMA (Hou et al. 2019) developed a cognitive-behavioural-therapy-based intervention program that used various techniques including reminder cards, jotting down thoughts and emotions before sleeping, and cognitive reconstruction. The results showed that compared to the control group, those who took part in the program saw improvements in their overall mental health, engagement in learning, sleep quality and also saw a drop in social media usage.

For specifically mobile applications, in 2013, a group of German researchers Löchtefeld, Böhmer, and Ganev (2013), developed an android-based mobile application called "AppDetox", where the main objective was to provide a "medication" for SA, with the app allowing users to set restriction-based rules on apps that prevent the user from opening the app at specific times of the day, until the user removed the rule, and after a set amount of time. It also displayed information on their set rules, times they've broken said rules, and usage statistics on all their apps. Their analysis found that users mostly set restrictions on messaging apps such as WhatsApp and WeChat, and broke self-created rules at roughly a 7:1 ratio (7 times a rules was broken for every 1 rule set).

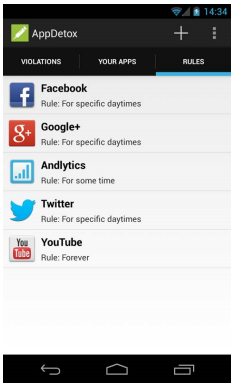

Figure 1: *Rules tab of AppDetox (Löchtefeld, Böhmer, and Ganev 2013)*

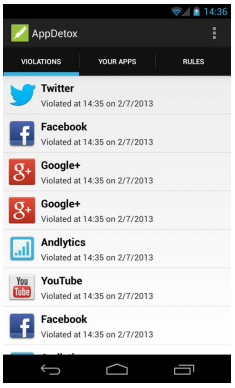

Figure 2: *Violations tab of AppDetox (Löchtefeld, Böhmer, and Ganev 2013)*

Another app titled "AntiSocial" developed by Australian company BugBean for Android devices provided similar functionality to AppDetox, allowing users to restrict access to certain apps and provided app usage statistics but also added other functionality such as showing the user the app usage of other users that were within their age group, location, and occupation. It also showed users how many times they picked up their device, and how long they used certain apps for (Michael 2017). According to the author of the journal entry, Katina Michael, the app ranked 3rd in downloads on the UK Google Play store mere days after launch, which "demonstrates the need for [AntiSocial.io]" (ibid.).

However, there seemingly exists a gap of intervention apps that deal specifically with SA or SMA, and that also allow self-reflection from the user. Most literature that's within the topic of intervention apps only provide analysis on apps that deal with adjacent issues. For example, a 2021 study by Oliveira et al. (2021) reviewed the effectiveness of psychological intervention mobile apps to help with issues such as depression, anxiety, and other related mental health issues, and while they concluded that the apps in the selected literature showed promise in reducing the aforementioned issues, none of the literature reviewed specifically looked at treating the issue of SA or SMA As well as this, most of these apps only provide the functionality to stop the user from using the app, but do not provide any sort of teachings for the user that can let them reflect on their usage.

1.2 Project Overview

1.2.1 Project Outline

Evidence has shown that smartphone apps that help with SA and SMA by restricting usage of applications are in-demand, and a substantial amount of research and studies have been conducted aiming to suggest treatments to the issue, however, very few apps have been researched, let alone developed and released with this approach in mind, with most consumer products on the app store opting to utilise app-restriction functionalities instead. Because of this approach, young adults who use these apps aren't given much opportunity to reflect on the nature of their addiction and the potential causes of it, should they choose not to seek outside professional assistance, leading to very little character growth for the end-user. However, by taking a different approach by integrating psychology for example, young adults can be given opportunity to reflect on their addiction and learn how to treat the issue. With that being said, the research statement is as follows:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help treat smartphone and social media addiction

1.2.2 Project Objectives

The primary aim of this project is to create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods. Two types of objectives have been identified, those being secondary objectives and primary objectives.

Secondary Objectives

Investigate existing mobile apps that use psychological methods to treat mental health issues
Research into how existing mobile applications make use of psychological methods when treating issues such as depression and anxiety to understand how viable using mobile apps are to treat similar issues.

Investigate the contextualisation of SA and SMA in existing apps
Research how the issue of SA and SMA are contextualised within existing intervention apps and what issues could arise from how they are presented. This will help in enabling growth in a user when it comes to learning about SA and SMA.

Investigate psychological methods used to treat smartphone and social media addiction
Research how psychological methods have been used in the past, specifically to treat SA or SMA, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate methods to track app usage
Research into how apps currently track phone activity, and how they can be utilised in the context of the project, as a major component of the application makes use of usage tracking.

Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained from the literature and technology review. (expand)

Gather participants for evaluating the app and perform evaluation
Participants will be used when evaluating if the developed application reduces smartphone and social media usage more than existing apps that use restriction functionality. They will be given an overview of the app before being given a consent form to sign to consent to participating in the research. Participants will be split into two groups, one will be using the developed app while the other will be using an existing app that has app restriction functionality.

Analyse the results and obtain a conclusion
Collect results from the evaluation and see if the app was more impactful at reducing smartphone usage than traditional apps. Then draw up a conclusion based on the results, and see if the problem statement was met.

1.3 Project Hypothesis

To test the hypothesis of this project, participants involved in testing the project will be split into two separate groups. One group will attempt to reduce their smartphone & social media usage by using a mobile application with usage tracking and psychological intervention methods, while the other will do the same, but by using a mobile application with traditional app restriction technologies instead. Given this, the hypothesis for this project is:

The combination of usage tracking and intervention methods rooted in psychology within a mobile application will help to reduce the smartphone & social media usage in young adults more than existing apps which only use app restriction technologies.

2 Literature and Technology Review

The literature and technology review is an integral component of the project as it allows the author to obtain foundational knowledge on the subject area that will then be used to carry out development of the primary research instrument. This is done by performing research and analysis into prior work surrounding the subject area.

This literature and technology review will review mobile apps that use psychological methods to treat mental health issues before pivoting into looking at how existing apps provide context to the issue of SA and SMA and the issues that may come with it. An investigation into how psychological methods can be used to treat smartphone and social media addiction is then performed before researching how app usage can be tracked.

2.1 Investigate existing mobile apps that use psychological methods to treat mental health issues

In the project background, it was briefly mentioned that although researchers have studied SA and SMA, few apps exist that tackle the issue are available, or been created for research purposes, though apps that treat similar issues exist and have extensively been reviewed and evaluated. Therefore this section aims to look at existing studies on apps to gain a further understanding on the viability of treating mental health issues.

2.1.1 Prior reviews on research-based mobile app interventions

German researchers Weisel et al. (2019) carried out a literature review of mobile apps that integrate psychological frameworks to treat mental health issues such as depression, anxiety, drug use, PTSD, sleep-related issues, and suicidal ideation, with their intention being to evaluate the effectiveness of these apps in reducing mental health symptoms.

The review discusses the kinds of components these apps utilise, with the most common being behaviour monitoring, participant engagement via explicit input, tailoring based on feedback, input, or sensing, and some form of gamification, which can be defined as "the idea of using game design elements in non-game contexts to motivate and increase user activity and retention" (Deterding et al. 2011). It also notes the psychological frameworks used, with a majority of apps opting to use CBT.

The review then details their findings, with them discovering that out of all the mental health issues targeted within the selected literature, only depression, sleep-related issues and smoking returned a "significant effect", with apps that targeted the other listed mental health issues failing to return significant effects, with the review stating that "our knowledge on how to design effective mental health apps is very much at the beginning" (Weisel et al. 2019).

The review discusses future research that should be investigated, such as different design and development approaches that focus more on the user and the framework, as well as engagement with the app via usage patterns, before concluding the review by saying that at the current point in time, there isn't much significant evidence that supports using mobile apps for treating mental health as an alternative to traditional treatment, highlighting that while apps targeting depression, sleep issues and smoking showed promising results, the results returned a high risk of bias within them.

Two years later, in 2021, Portuguese researchers Oliveira et al. (2021) carried out a literature review of mobile app-based interventions utilising psychological methods that targeted college and university students, aiming to review the outcomes of using app-based psychological interventions as opposed to traditional therapy, as well as looking how these apps are being embraced by students.

After an extensive literature selection process, the review then details the types of mental health issues these apps target such as anxiety, depression and stress, while also noting that a majority of these apps are "self-guided" in nature, (ibid.) with them looking to prevent the issue. It then details the psychological frameworks the apps use, with cognitive-behavioural therapy (CBT) being the main framework used or combined with another framework, while also mentioning other frameworks such as Acceptance and Commitment Therapy (ACT) and Self-determination Theory (SDT).

The review then explores the outcomes of the intervention apps and their associated results. It then discusses the selected literature as a whole, mentioning how apps that target a specific mental health issue such as anxiety depression, and stress showed promise in mitigating symptoms associated with those issues. It also states that most of the apps showed indications of being well constructed at the conceptual level according to all available evidence, and that the apps returned promising results in regards to student acceptability. One interesting aspect the review notes is that privacy is rarely, if ever, brought up when discussing the design and development of these apps, with the study emphasising that "best practices should be known and shared" (ibid.), implying that the privacy of an end-user isn't considered

a major factor when developing these apps, a sentiment that is also echoed by similar papers, such as Marshall, Dunstan, and Bartik (2019) who mentions it in regards to users not knowing if their personal info is being used, though this may be due to the apps being developed for research purposes, and not intended to be used by a wider audience.

The review then concludes by saying that the current literature demonstrates that using mobile apps for mental health intervention is a viable option and that using these intervention apps could be a great resource for counselling services. It notes however that future work into the topic should utilise alternative psychological methods, as well as looking further into what students and therapists would want out of similar apps, and that future studies should stray from monetary incentives for students looking to participate in said studies (Oliveira et al. 2021).

2.1.2 Conclusion

To conclude this section, the first review analysed available studies which covered developing a mobile app that integrated psychological frameworks to target mental health issues, and found that while apps that targeted depression, sleep-related issues, and smoking showed some results, the field was merely in its infancy when it came to designing effective applications. Then, a similar review performed later in time showed that mobile apps that targets mental health issues aimed towards college-level students were a viable alternative option for treating mental health issues due to student embracement, showcasing growth within the topic between the two points in time, though noted that this could've been due to incentives attached to using the apps. This as well as the fact that these studies lack any insight on user privacy or psychological frameworks other than mostly CBT, show a need to expand the types of psychological frameworks used for these apps, provide more clarity on user privacy when developing the app, and stray away from incentives for a user which may skew the results of user retention. Overall though, the above reviews demonstrate that the viability of a mobile mental health app that uses psychological frameworks has grown over time, and that developing such app is doable, as long as aspects such as other frameworks, user privacy, and user incentive are considered.

2.2 Investigate the contextualisation of SA and SMA in existing apps

Near the end of the project background, it was mentioned that because of how intervention apps are developed with mostly functionality in mind, while neglecting

any sort of teaching element, therefore leaving the user little to no room to learn about SA and SMA. This section intends to review how SA and SMA are currently contextualised, and how this could be improved.

2.2.1 Current contextualisation, or Past contextualisation issues (temp)

Researcher Dosselaar (2017) put forward an argument stating that SA apps created by developers aren't suitable for treating SA due to developers having limited knowledge on the topic of SA, and points out the irony in attempting to treat SA with the use of a Smartphone. This sentiment is also shared within the book Petrucco (2021), stating that proposing technology as a solution to an issue where the cause is the technology itself is paradoxical in nature.

The author goes into extensive detail about the many issues with the concept of SA apps, including how because the topic of SA is being solved by developers instead of wider society, they lack a deep understanding of the topic due to these apps being creating with profits in mind, as well as pointing out the overall complexity of the issue, mentioning how SA is a product of many other factors; not just the phone, and with the creation of these apps, they simplify what is often a complex issue by acting as a one-size-fits-all solution (Dosselaar 2017). This sentiment is shared by another study by Roffarello and Russis (2019) who notes that users of these apps can be aware that they aren't sufficient solutions on their own.

One of the major problems with these apps the author notes is how they lack any kind of opportunity for the user to understand and reflect on the nature of SA. In particular, the author notes that these apps frame SA as an issue that can be easily measured, while lacking any features that offer information about SA. A separate study mentions how opportunities to learn about SA play a key role in solving the issue (Rosa, Stracuzzi, and Faro 2022).

The author also notes that these apps also fail to consider context behind smartphone usage, mentioning how "smartphone usage can be productive", and a lack of usage can be unproductive (Dosselaar 2017). Petrucco (2021) backs this argument by saying using SNS on public transit is different from using it in a class.

The author concludes the research portion by stating that SA apps are limited in effectiveness as they focus on practical functionality of the app rather than the contextualisation and opportunity for reflection for the user regarding SA. Rahmillah et al. (2023) provides a similar statement, stating that characteristics such as knowledge of their relationship between their smartphone and overall health, among others, would influence app effectiveness. The author then recommends that to properly solve the issues of SA, other aspects such as the design of the apps and other psychological frameworks should be looked into (Dosselaar 2017).

2.3 Investigate psychological methods used to treat smartphone and social media addiction

One part of the project background mentioned that plenty of studies have been conducted utilising different psychological methods to treat SA and SMA. This section aims to review the most common psychological methods to understand which method would work best with the intention of the project.

2.3.1 section to be named

review goes here

2.4 Investigate methods to track app usage

The project background mentioned that existing SA and SMA intervention applications mainly make use of app-restriction and usage tracking technologies, and due to usage tracking being a large component of the project, it would be useful to know the technological methods used for app usage tracking.

2.4.1 Justification for app usage tracking

In some apps, app usage tracking has been implemented and shown to be successful in reducing smartphone usage, a study by researchers Rahmillah et al. (2023) review the effectiveness of apps intended to reduce Smartphone use, and found that in the 13 apps reviewed, 4 apps successfully reduced smartphone usage, those of which included app usage tracking, though it should be noted that all apps included some form of usage tracking.

2.4.2 Methods of app usage tracking

A study by Rosa, Stracuzzi, and Faro (2022) hypothesised that individuals with high levels of smartphone usage show more difficulty in not using their smartphone, when compared to those with lower levels. The researchers carried out their verification of the hypothesis by developing an application called "SocialStatsApp" to track app usage of participants, alongside the use of questionnaires such as the Smartphone Addiction Scale-Short Version, a questionnaire used to screen adolescents who may be vulnerable to smartphone addiction (Kwon et al. 2013), as well as cognitive tests using external platforms.

The app was developed for devices using the Android operating system, Google's own mobile phone OS. The researchers mention how making the app compatible for iOS devices, Apple's mobile phone OS, was not possible, as the OS disallows the

collection of usage statistics for privacy reasons (Apple-Support n.d.). The programming language used was Java, one of three officially supported languages via the Android SDK, the other two being Kotlin and C++ (Android-Developers n.d.), though the study doesn't detail the reasoning behind using Java. For usage tracking, the app makes use of the Android class android.app.usage.UsageStatsManager from the Android API which gives access to phone usage stats and history. Because of this, the app needs permission manually granted by the user.The results of the study showed that those with high levels of smartphone usage did experience more difficulty in not using their phone.

Researchers Bye et al. (2024) undertook a study analysing the effects technology has on the mental health of young people, and part of their research included developing an app called the "3S-YP" app that would constantly collect what the researchers vaguely refer to as "smartphone metadata" from participants. While they don't detail the development of the app, they do mention that smartphone metadata was collected using the android.app.usage package from the Android API, while also mentioning that metadata collection was not performed for the iOS version of the application due to "data protection restrictions", a sentiment shared by the previous study, showing that developing for Android allowed for additional functionality not possible with iOS.

Outside of using the Android API, another method of tracking a user's app usage is by making use of screenshots and digital well-being features bundled with smartphones. Ubochi (2019) investigates the use of digital health apps and their effectiveness via Apple's Screen Time feature. Due to the aforementioned privacy restrictions apple places on third-party apps when it comes to collecting usage data, the researcher instructs participants to instead screenshot multiple aspects related to the Screen Time feature. A limitation of this method however is that screenshots must be taken at a specified time, as mistiming it could yield less accurate app usage (*Mobile sensing in psychology: Methods and applications* 2024).

2.4.3 Discussion

Both studies chose Android as their development platform of choice, mentioning that iOS apps don't allow for app usage tracking due to privacy restrictions (Rosa, Stracuzzi, and Faro 2022; Bye et al. 2024; Apple-Support n.d.). This is elaborated further in the book *Mobile sensing in psychology: Methods and applications* (2024) that mentions include iOS users would require a "jailbroken" phone, which brings a variety of issues with it. Meanwhile, Android allows for this collection of data, as long as the user permits it. Only one of the studies makes mention of a programming language (Rosa, Stracuzzi, and Faro 2022), that being Java, and while both studies

make use of the android.app.usage package within the Android API, only one (Rosa, Stracuzzi, and Faro 2022) explicitly uses the UsageStatsManager class. Given this, it is clear that the OS that the app should be developed for is Android, and that using the android.app.usage package would be a viable method of app usage tracking.

3 Project's Primary Research Methods

This section will give a detailed overview on the methods that will be used to carry out the remainder of the project and provide a reasonable conclusion to the problem statement and hypothesis. This section will also outline any possible ethical issues, what work has been completed up to this point and all remaining work that will be carried out over the project's lifetime.

3.1 Research Method and Justification

3.2 Develop the Mobile app (temp)

3.2.1 Development

3.2.2 Project Lifecycle

3.3 Usability Testing with users and Evaluation

3.3.1 Usability Testing

3.3.2 Evaluation

3.4 Ethical Issues

3.5 Preliminary Work

3.6 Remaining Work

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